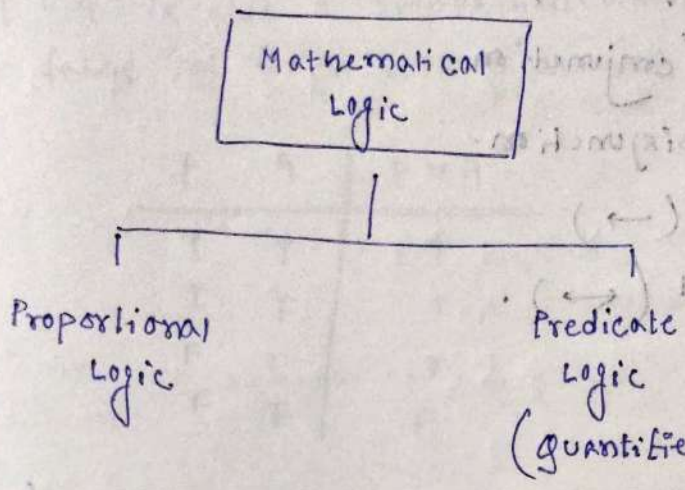
Discrete Mathematics

Logic



- A declorative sentence to which we can assign only one of the truth values (i.e. True / False) is called a proportional statement or proportion.

Ex. India's capital is Kolkata : False
India's capital is Delhi : True.

- **Assumptions of proportion :**

Law of excluded middle : If a proposition is not true then it will be false and similarly if a proportion is not false then it will be true.

Law of contradiction : A proportion can't be true as well as false simultaneously.

- **Atomic propositions :** Atomic propositions are simple proportions which can't be divided further.

- **Compound propositions :** Compound propositions are constructed by combining atomic propositions with the help of connectives. Compound propositions are also known as propositional formula or proportional function.

Connectives:

1. Negation ($\neg$ / $\sim$) / Not.
2. And ($\wedge$) / conjunction.
3. OR ($\vee$) / Disjunction.
4. Implication ($\rightarrow$)
5. Biconditional ($\leftrightarrow$).

1. Negation:

If p is any proposition, then negation of p / not(p) / $\sim p$ is also a proposition whose truth value is true only when p is false and it is false only when p is true.

p	$\sim p$
T	F
F	T

2. AND ($\wedge$) / conjuction:

If p and q are any two propositions, then "p AND q / $p \wedge q$" is also a proposition whose truth value is True only when both p as well as q are true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

3. OR ($\vee$) / Disjunction:

If p and q any two propositions, then 'p or q' / $p \vee q$ is also a proposition whose truth value is false only when both p as well as q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

4. Implication ($\rightarrow$):

If p and q are any two propositions, then p implies q (i.e. $p \rightarrow q$ / if p then q) is also a proposition whose truth value is false only when p is true and q is false.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$p \rightarrow q \equiv \sim p \vee q$

p	q	$\sim p$	$p \rightarrow q$	$\sim p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$\therefore$ verified.

$q \to p \equiv$ [illegible] $\sim p \to \sim q$

Implications:

1. If p is true, then q must be true for $p \to q$ to be true.
2. If p is false, then $p \to q$ is always true whatever is the truth value of q.
3. If q is true, the $p \to q$ is always true whatever is the truth table of p.
4. Truth table of $p \to q$ and truth table of $\sim p \vee q$ are exactly same, therefore both are equivalent
5. converse of $p \to q$ is $q \to p$.
6. opposite of inverse of $p \to q$ is $\sim p \to \sim q$.
7. contra positive for $p \to q$ is $\sim q \to \sim p$.

Biconditional ($\leftrightarrow$):

If p and q are any two propositions, then $p \leftrightarrow q$ (i.e. p if and only if q) is also a proposition whose truth value is true only when both p and q have same truth value.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

∴ verified.

- Tautology:

A propositional function which is always true is called tautology.

A propositional function which is always true is also called valid propositional function.

Ex.

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

∴ $p \vee \sim p$ is a tautology.

p	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

∴ $p \rightarrow (q \rightarrow p)$ is a tautology.

How many propositional functions can be defined 'n' propositional variable?

using

→ 2^{2^n}.

Contradiction:

A propositional function which is always false is called a contradiction.

If a propositional functim is false for atleast one case is called a invalid tonclim.

Contingency:

A propositional function which is neither a tautology nor a contradiction is called a contingency.

i.e. A contingency is true for at least one case as well as false for atleast one case.

Satisfiable:

A propositional function which is true for at least one case is called satisfiable propositimal function.

Note:

1. Every propositional function which is not contradiction is satisfiable.
2. Every tautology is satisfiable but not vice versa.
3. Every contingency is satisfiable. but vice versa not true.

Logical implication :

Let p and q are any two propositional functions.

whenever p is true if q is true then p logically implies q. is true.

If there exist any case for which p is true but q is false. then p logically implies q. is invalid.

p logically implies q. if and only if $p \rightarrow q$ is a tautology.

Logical equivalence :

1. If p and q are any two propositional functions. then p equivalent to q is written as $p \equiv q$.
2. p and q are said to be equivalent if and only if they have same truth table.
3. $p \equiv q$ if and only if $p \leftrightarrow q$ is tautology.

Algebric laws of connectives / some important equivalences :

1. $p \vee p \equiv p$ & $p \wedge p \equiv p$ [Idempotent law].
2. $(p \vee q) \vee R \equiv p \vee (q \vee R)$
   $(p \wedge q) \wedge R \equiv p \wedge (q \wedge R)$ [Associative]
3. $p \vee q \equiv q \vee p$
   $p \wedge q \equiv q \wedge p$ [commutative]
4. $p \wedge (q \vee R) \equiv (p \wedge q) \vee (p \wedge R)$
   $p \vee (q \wedge R) \equiv (p \vee q) \wedge (p \vee R)$ [Distributive].

5. $\sim(p \vee q) \equiv \sim p \wedge \sim q$ [De Morgan's]

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

6. $p \wedge (p \vee q) \equiv p$ [Absorptive law].

$p \vee (p \wedge q) \equiv p$

7. $p \vee (\text{contradiction}) \equiv p$

$p \vee (\text{Tautology}) \equiv \text{Tautology}$.

8. $p \wedge (\text{Tautology}) \equiv p$

$p \wedge (\text{contradiction}) \equiv \text{contradiction}$

9. $p \vee (\sim p) \equiv \text{Tautology}$.

$p \wedge (\sim p) \equiv \text{contradiction}$.

10. $\sim(\text{contradiction}) \equiv \text{Tautology}$

$\sim(\text{Tautology}) \equiv \text{contradiction}$.

11. $\sim(\sim p) \equiv p$.

■ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

p	q	r	$q \vee r$	$p \wedge (q \vee r)$ (LHS)	$p \wedge q$	$p \wedge r$	Result (RHS)
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Some important equivalences:

1. P implies Q : $P \rightarrow Q$.
2. If P then Q : $P \rightarrow Q$.
3. P only if Q : $P \rightarrow Q$.
4. P is sufficient condition for Q : $P \rightarrow Q$.
5. Q is necessary condition for P : $P \rightarrow Q$.
6. P if Q : $Q \rightarrow P$
7. P when Q : $Q \rightarrow P$
8. P follows from Q : $Q \rightarrow P$.
9. P unless $\sim Q$: $\sim Q \rightarrow P$
10. P unless Q : $\sim Q \rightarrow P$.

Prove that. $(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$ is a tautology.

P	Q	R	$(P \rightarrow Q)$	$(Q \rightarrow R)$	$(P \rightarrow R)$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$	Final
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

∴ For all values the result is true

So, it is a tautology.

Show that, $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$(p \wedge q) \rightarrow (p \vee q) \equiv \sim(p \wedge q) \vee (p \vee q)$

$[\because p \rightarrow q \equiv \sim p \vee q]$

$\equiv (\sim p \vee \sim q) \vee (p \vee q)$ [De Morgan's]

$\equiv ((\sim p \vee \sim q) \vee p) \vee q$ [Associative]

$\equiv ((\sim p \vee p) \vee (\sim q \vee p)) \vee q$ [Distributive]

$\equiv ((\text{tautology}) \vee (\sim q \vee p)) \vee q$ [complement]

$\equiv (\sim q \vee p) \vee q$ [Identity]

$\equiv (p \vee \sim q) \vee q$ [commutative]

$\equiv p \vee (\sim q \vee q)$ [Associative]

$\equiv p \vee (\text{tautology})$ [complement]

$\equiv$ tautology [identity]

Without using truth table prove the following equivalence:

$(q \rightarrow (p \wedge \sim p)) \rightarrow (r \rightarrow (p \wedge \sim p)) \equiv r \rightarrow p$.

LHS: $(q \rightarrow (p \wedge \sim p)) \rightarrow (r \rightarrow (p \wedge \sim p))$

$\equiv (q \rightarrow \text{contradiction}) \rightarrow (r \rightarrow \text{contradiction})$

$\equiv (\sim q \vee \text{contradiction}) \rightarrow (\sim r \vee \text{contradiction})$

$\equiv (\sim q \rightarrow \sim r)$

$\equiv q \vee \sim r \equiv \sim r \vee q \equiv r \rightarrow q$ [Proved]

Let p : He is intelligent.
q : He is tall

Sc two propositions. write each of the following statements using symbolic form using p and q :

1. He is tall but not intelligent.
2. He is neither tall nor intelligent.
3. He is intelligent or he is tall.
4. It is not true that he is intelligent or tall.
5. It is not true that he is not tall or not intelligent.

Soln : 1. $p \wedge \sim q$.

2. $\sim p \wedge \sim q$.

3. $p \vee q$.

4. $\sim (p \vee q)$.

5. $\sim (\sim p \vee \sim q)$.

Show that. $(\sim q \wedge (p \to q)) \to \sim p$ is a tautology.

p	q	$\sim q$	$p \to q$	$\sim q \wedge (p \to q)$	$\sim p$	Final
T	T	F	T	F	F	T
T	F	T	F	F	F	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T

So, it is a tautology.

$(\sim q \wedge (p \to q)) \to \sim p$

$\equiv (\sim q \wedge (\sim p \vee q)) \sim [p \to q \equiv \sim p \vee q]$
$\sim p$

$\equiv \sim(\sim q \wedge (\sim p \vee q)) \vee \sim p$

$\equiv \sim(q \vee \sim(\sim p \vee q)) \vee \sim p$

$\equiv q \vee (p \wedge \sim q) \vee \sim p$

$\equiv \sim((\sim q \wedge \sim p) \vee (\sim q \wedge q)) \vee (\sim p)$ [Distributive]

$\equiv \sim(\sim(q \vee p) \vee \text{contradiction}) \vee (\sim p)$

$\equiv ((q \vee p) \wedge \sim(\text{contradiction})) \vee (\sim p)$

$\equiv ((q \vee p) \wedge \text{tautology}) \vee (\sim p)$

$\equiv (q \vee p) \vee (\sim p)$

$\equiv q \vee (p \vee \sim p)$

$\equiv q \vee \text{tautology}$.

$\equiv$ tautology. [Proved].

check wheather the statement is a tautology or contradiction using truth table:
without.

$p \to (q \to p)$.

$p \to (q \to p)$

$\equiv p \to (\sim q \vee p)$

$\equiv \sim p \vee (\sim q \vee p)$

$\equiv (\sim p \vee p) \vee \sim q$

$\equiv$ tautology $\vee \sim q$

$\equiv$ tautology.

Show that. $(p \vee q) \wedge (\sim p \wedge \sim q)$ is contradiction:

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim p \wedge \sim q$	Final
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

All the values of $(p \vee q) \wedge (\sim p \wedge \sim q)$ are false.
So. it is a contradiction.

Prove that. among inverse, converse and contrapositive of the statement $p \to q$ only contrapositive is equivalent to $p \to q$.

p	q	$\sim p$	$\sim q$	$p \to q$	$q \to p$	$\sim p \to \sim q$	$\sim q \to \sim p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

$(\sim q \to \sim p) \equiv (p \to q)$.

show that. $p \wedge q$ logically implies $p \leftrightarrow q$.

p	q	$p \wedge q$	$p \leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	T

→ This shows that $(p \wedge q)$ logically implies $p \leftrightarrow q$.

Argument / Inference:

The statement that, "A set of premises $\{p_1, p_2, p_3, \ldots, p_n\}$ yields another proposition q" is called an argument.

q is called conclusion of the argument.

Ex.

1. If today is Radha's Birthday, then today is 15th march.
2. Today is 15th march.

∴ Today is Radha's birthday.

— The conclusion is not garanteed to be true.

So, the statement is invalid.

1. Argument may be valid or invalid.
2. The process of reasoning whether the argument is valid or invalid is called inference.
3. If conclusion Q can be inferred from the set of premises by applying some rules of inference and equivalences, then argument is said to be valid otherwise invalid.

Following statements are valid:

1. Argument $\{P1, P2, P3, \ldots, Pn\} \vdash Q$ is valid.
2. $\{P1, P2, P3 \ldots Pn\}$ logically implies Q is true / valid.
3. $\{P1 \wedge P2 \wedge P3 \ldots \wedge Pn\} \rightarrow Q$ is a tautology

■ Rules of Inference:

Inference is the process of deriving logical conclusions from certain basic assumptions, called premises by employing certain principles of reasoning called rules of inference. Any valid reasoning is rule of inference.

To check the validity of an argument we shall frequently use three basic rules of inference called rules of P, T and CP as given below -

Rule P: A premises may be introduced at any step in the derivation.

Rule T: A formula S may be introduced in the derivation if S is tautologically implied by any one or more preceding formulas in the derivation.

Rule CP: If we can derive a formula s from another formula r and a set of premises, then the statement $r \to s$ can be derived from the set of premises alone.

1. Simplification: $P \wedge Q \equiv T$ / $\therefore P$ is true, } valid argument

Similarly, $P \wedge Q \equiv T$ / $\therefore Q$ is valid }

- $(P \wedge Q)$ logically implies P is valid.

$(P \wedge Q) \to P$ is a tautology.

2. Addition: P / $\therefore P \vee Q$ } is valid

P logically implies $(P \vee Q)$ is valid.

$P \to (P \vee Q)$ is a tautology.

- P / $\therefore P \wedge Q$ } is invalid.

3. **Conjunction:**

$$\left.\begin{array}{l} P \\ Q \\ \therefore P \wedge Q \end{array}\right\} \text{ is valid.}$$

4. **Disjunctive syllogism:**

$$\left.\begin{array}{l} P \vee Q \\ \sim P \\ \therefore Q \end{array}\right\} \text{ is valid.}$$

$$\left.\begin{array}{l} P \vee Q \\ \sim Q \\ \therefore P \end{array}\right\} \text{ is valid.}$$

5. **Conjunctive syllogism:**

$$\left.\begin{array}{l} \sim(P \wedge Q) \\ P \\ \therefore \sim Q \end{array}\right\} \text{ is valid.}$$

$$\left.\begin{array}{l} \sim(P \wedge Q) \\ Q \\ \therefore \sim P \end{array}\right\} \text{ is valid.}$$

6. Modus Ponens :

$P \rightarrow Q$
P
$\therefore Q$

} is valid.

7. Fallacy of affirming the consequent :

$P \rightarrow Q$
Q
$\therefore P$

} is invalid.

8. Modus Tollen's :

$P \rightarrow Q$
$\sim Q$
$\therefore \sim P$

} is valid.

9. Fallacy of denying the antecedent :

$P \rightarrow Q$
$\sim P$
$\therefore \sim Q$

} is invalid.

10. Transitivity:

$P \rightarrow Q$
$Q \rightarrow R$
$\therefore P \rightarrow R$

} is valid.

11. Dilemma:

$P \rightarrow R$
$Q \rightarrow R$
$P \vee Q$
$\therefore R$

} is valid.

12. Constructive Dilemma:

$P \rightarrow R$
$Q \rightarrow S$
$P \vee Q$
$\therefore R \vee S$

} is valid.

13. Destructive Dilemma:

$P \rightarrow R$
$Q \rightarrow S$
$\sim R \vee \sim S$
$\therefore \sim P \vee \sim Q$

} is valid.

14. Resolution:

$$\left.\begin{array}{l} P \vee Q \\ \sim P \vee R \\ \therefore Q \vee R \end{array}\right\} \text{ is valid.}$$

15. Non Sequitor:

$$\left.\begin{array}{l} P \\ Q \\ \hline \therefore R \end{array}\right\} \text{ is invalid.}$$

Let p denote "A person is rich" and q denote "A person is happy". Write each statement in symbolic form using p and q:

(1) If he is rich then he is unhappy.

(2) He is neither rich nor happy.

(3) It is necessary to be poor in order to be happy.

(4) To be poor is to be happy.

(5) He is rich but unhappy.

(i) $p \rightarrow \sim q$.

(ii) $\sim p \wedge \sim q$.

(iii) $q \rightarrow p$

(iv) $\sim p \leftrightarrow \sim q$.

(v) $p \wedge \sim q$.

p: A person is rich.

q: A person is happy.

Determine whether the following argument is valid or invalid:

"If Amy gets the correct answer, then Ben won't. If Ben gets the correct answer, then Amy won't. Therefore, either Amy or Ben will get the correct answer."

P: Amy gets the correct answer.

q: Ben gets the correct answer.

1. $p \rightarrow \sim q$.
2. $q \rightarrow \sim p$.
3. $\therefore p \vee q$.

This will be valid if $P_1 \wedge P_2 \rightarrow Q$ is a tautology.

$$\therefore \{\underbrace{(p \rightarrow \sim q)}_{P_1}, \underbrace{(q \rightarrow \sim p)}_{P_2}\} \vdash \underbrace{(p \vee q)}_{Q}$$

p	q	$\sim p$	$\sim q$	$p \rightarrow \sim q$	$q \rightarrow \sim p$	$p \vee q$	$P_1 \wedge P_2$	$P_1 \wedge P_2 \rightarrow Q$
T	T	F	F	F	F	T	F	T
T	F	F	T	T	T	T	T	T
F	T	T	F	T	T	T	T	T
F	F	T	T	T	T	F	T	F

$\therefore$ Argument is not a valid argument.

Verify the validity of the argument:

"If I drive to work, then I will arive in time.
I don't drive to work.
Therefore I will not arrive in time."

p : I drive to work

q : I will arrive in time.

The symbolical form of the given argument.

$p \to q$

$\sim p$

$\therefore \sim q$

To check the ~~validity~~ validity of the argument we check $((p \to q) \wedge \sim p) \to \sim q$ is a tautology.

p	q	$\sim p \vee \sim q$	$p \to q$	$(p \to q) \wedge \sim p$	R	$R \to \sim q$
T	T	F	F	T	F	T
T	F	F	T	F	T	T
F	T	T	F	T	T	F
F	F	T	T	T	T	T

This is not a tautology

Hence the argument is valid.

Show that s is valid conclusion from the premises

$p \to \sim q,\ q \vee r,\ \sim s \to p,\ \sim r$

The steps for deducting wheather s is a valid conclusion from the given premises are given below.

Step 1: $q \vee r$

Rule P / permise.

Step 2: $\sim r$

Rule P / Permise.

step 3: q

Steps 1, 2 and by using disjunctive syllogism.

step 4: $p \to \sim q$.

Rule P.

Step 5: $\sim(\sim q)$

Rule T and using $\sim(\sim q) \equiv q$.

step 6: $\sim p$

Step 4, 5 & by Modus Tollens.

Step 7: $\sim s \to p$

Rule P.

step 8: $\sim(\sim s)$.

Step 6, 7 & by Modus Tollens.

Step 9: s

Rule T & using $\sim(\sim s) \equiv s$

Thus, s is a valid conclusion from given premises.

Show that t is a valid conclusion from the premises.

$p \to q,\ q \to r,\ r \to s,\ \sim s$ and $p \vee t$.

Step	Statement	Reason
1.	$p \to q$	Rule P
2.	$q \to r$	Rule P
3.	$p \to r$	Step 1.2 & using transitivity.
4.	$r \to s$	Rule P
5.	$p \to s$	Steps 3.4 & using transitivity
6.	$\sim s$	Rule P.
7.	$\sim p$	Step 5.6 & Modus Tollens
8.	$p \vee t$	Rule P.
9.	t	Steps 7.8 & by using disjunctive syllogism.

Hence, t is the valid ~~arg~~ conclusion

Represent the argument

If I work hard then I will get success.

I work hard

I got success.

p : I work hard

q : I will get success

$p \to q$

p

$\therefore q$

∴ The argument is :

$(p \to q), p) \vdash q$.

This will be valid if $(p \to q) \wedge p \to q$ is a tautology.

We shall prove this using laws of proposition.

$(p \to q) \wedge p \to q$

$\equiv \sim((p \to q) \wedge p) \vee q \quad [\because p \to q \equiv \sim p \vee q]$

$\equiv \sim((\sim p \vee q) \wedge p) \vee q$

$\equiv (\sim(\sim p \vee q) \vee \sim p) \vee q \quad [\text{De Morgan's law}]$

$\equiv ((p \wedge \sim q) \vee \sim p) \vee q \quad [\text{D. Morgan's}]$

$\equiv ((p \vee \sim p) \wedge (\sim q \vee \sim p)) \vee q \quad [\text{Distributive}]$

$\equiv (T \wedge (\sim q \vee \sim p)) \vee q$

$\equiv (\sim q \vee \sim p) \vee q$

$\equiv (\sim p \vee \sim q) \vee q \quad [\text{commutative}]$

$\equiv \sim p \vee (\sim q \vee q) \quad [\text{commutative}]$

$\equiv \sim p \vee T$

$\equiv T \quad (\text{tautology})$.

∴ The argument is valid.

Express the following argument symbolically and test wheather this is valid argument:

If it rains today then cricket match will not be played.

Cricket match is not played.

Therefore, it rained today.

p : It rains today.

q : Cricket match is not played.

The symbolical form of the given argument

$$p \rightarrow \sim q$$

$$\sim q$$

$$\therefore p$$

The argument is $((p \rightarrow \sim q) . q) \vdash p$

To be $(p \rightarrow \sim q)$ true when q is already true, p is not guaranteed to be true.

$\therefore$ The argument is not valid.

Check wheather this is a valid argument:

$\{p \rightarrow q . \sim(p \wedge q)\}$ logically implies q.

p	q	$p \rightarrow q$	A: $p \wedge q$	B: $\sim(p \wedge q)$	C: $A \wedge B$	$C \rightarrow q$
T	T	T	T	F	F	T
T	F	F	F	T	F	T
F	T	T	F	T	F	T
F	F	T	F	T	F	T

Normal Forms

Conjuctive Normal Form (CNF):

A compound proposition P is said to be conjuctive normal form (CNF) of r simple propositions $P_1, P_2 \ldots P_r$ if the expression P can be expressed as $P = P_1 \wedge P_2 \wedge \ldots \wedge P_r$ where each P_i = disjunction of a finite no. of the simple propositions of each of $P_1, P_2 \ldots P_r$ or their negations and $P_i \neq P_j$ for all distinct i and j.

Illustrations:

(i) $P = (\sim P_1 \vee P_3 \vee P_2) \wedge (\sim P_1 \vee \sim P_2 \vee \sim P_3) \wedge (P_1 \vee P_2 \vee P_3)$ is a CNF of the simple propositions P_1, P_2, P_3.

But $P = (\sim P_1 \vee P_2) \wedge (\sim P_1 \vee \sim P_2 \vee \sim P_3)$ is not a full CNF since the expression $(\sim P_1 \vee P_2)$ does not contain P_3 or its negation.

(ii) $P = P \vee (\sim P \wedge q)$ is not a CNF

(iii) $(\sim p \vee q) \wedge (\sim p \vee q)$ is a CNF of the two simple propositions p and q as well as full CNF.

Disjunctive Normal Form (DNF):

A compound proposition P is said to be DNF of r simple propositions $P_1, P_2 \ldots P_r$ if the expression P can be expressed as $P = P_1 \vee P_2 \vee \ldots \vee P_s$, where each P_i = Conjunction of a finite no. of the simple propositions of each $P_1, P_2, \ldots, P_r$ or their negations and $P_i \neq P_j$ for all distinct i & j.

Illustration:

1. $P = (p \wedge \sim r) \vee (\sim q \wedge p \wedge r) \vee (\sim p \wedge q)$ is a DNF.
2. $(p \wedge \sim r) \vee (p \wedge \sim q \wedge r) \vee (\sim r \wedge p \wedge q)$ is a DNF

Write a CNF of the following statements:

1. $p \wedge (p \rightarrow q)$
2. $\{q \vee (p \wedge r)\} \wedge \sim\{(p \vee r) \wedge q\}$

(1) $p \wedge (p \rightarrow q)$

$\equiv p \wedge (\sim p \vee q)$

– which is the required CNF.

(2) $\{q \vee (p \wedge r)\} \wedge \sim\{(p \vee r) \wedge q\}$

$\equiv \{q \vee (p \wedge r)\} \wedge \{\sim(p \vee r) \vee \sim q\}$ [D. Morgan's]

$\equiv \{q \vee (p \wedge r)\} \wedge \{(\sim p \wedge \sim r) \vee \sim q\}$

$\equiv (q \vee p) \wedge (q \vee r) \wedge (\sim p \vee \sim q) \wedge (\sim r \vee \sim q)$

[Distributive law]

– which is the required CNF.

write the DNF of the following statement:

$$P \rightarrow (P \wedge (Q \rightarrow P)).$$

$P \rightarrow (P \wedge (Q \rightarrow P))$

$\equiv \sim P \vee (P \wedge (Q \rightarrow P))$

$\equiv \sim P \vee (P \wedge (\sim Q \vee P))$

$\equiv (\sim P \vee P) \wedge (\sim P \vee (\sim Q \vee P))$ [Distributive law]

$\equiv T \wedge (\sim P \vee (\sim Q \vee P))$ [Complement]

$\equiv \sim P \vee (\sim Q \vee P)$ [Identity law]

$\equiv \sim P \vee (P \vee \sim Q)$ [commutative law]

$\equiv (\sim P \vee P) \vee \sim Q$ [Associative law]

$\equiv T \vee \sim Q$ [complement law]

$\equiv T$

$\equiv P \vee \sim P$

– which is a required DNF.

write DNF of the following statement.

① $\sim \{\sim (P \leftrightarrow Q) \wedge R\}$

$\equiv \sim \{\sim ((P \rightarrow Q) \wedge (Q \rightarrow P)) \wedge R\}$

$\equiv \sim \{\sim ((\sim P \vee Q) \wedge (\sim Q \vee P)) \wedge R\}$

$\equiv \sim \{\sim (\sim P \vee Q) \vee \sim (\sim Q \vee P) \wedge R\}$.

$\equiv (\sim P \vee Q) \wedge (\sim Q \vee P) \vee \sim R$

$\equiv (\sim P) \vee (Q \wedge \sim Q) \vee P \vee (\sim R)$

– which is required DNF.

(11) $\{p \wedge \sim(q \wedge r)\} \vee (p \to q)$

$\equiv \{p \wedge \sim(q \wedge r)\} \vee (\sim p \vee q)$

$\equiv \{p \wedge (\sim q \vee \sim r)\} \vee (\sim p \vee q)$ [De Morgan's]

$\equiv \{(p \wedge \sim q) \vee (p \wedge \sim r)\} \vee (\sim p \vee q)$ [Distributive law]

$\equiv (p \wedge \sim q) \vee \{(p \wedge \sim r) \vee (\sim p \vee q)\}$

$\equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee \sim p \vee q$ [Associative]

- which is the required DNF

Quantification and Quantifier:

We have seen a proposition can be created from a propositional function $P(x)$ by assigning particular values to the variable x.

There are two types of quantifications, namely Universal Quantifications and Existential Quantifications.

Universal Quantification: The universal quantification of the propositional function $P(x)$ is the proposition "$P(x)$ is true for all values of x in the domain". This quantification is denoted by $\forall x, P(x)$. The symbol $\forall$ is called universal quantifier.

1. Consider the propositional function $P(x)$ stating "x is a good batsman" where the domain is set of all students in a class. There its universal quantification.

   $\forall x, P(x)$ is the statement "Every student in the class is a good batsman.".

2. Let, $P(x)$ = "x is mortal" be a propositional function. then its universal quantification $\forall x, P(x)$ is the proposition "All human beings are mortal", where the domain is the set of all human beings. Obviously this proposition $\forall x, P(x)$ is true.

Existential Quantification:

The existential quantification of the propositional function $P(x)$ is the proposition "there exists a value of x in the domain for which $P(x)$ is true." This quantification is denoted by $\exists x, P(x)$. The symbol $\exists$ is called Existential Quantifier.

1. Let $P(x)$ be the propositional function "$x+4 < 15$". Then its existential quantification $\exists x, P(x)$ is "there exists a value of x for which $x+4<15$ is valid" where the domain is the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$. Obviously the proposition $\exists x, P(x)$ is true. But it would be false if the domain is taken as the set $\{12, 13, 14, 15\}$.

2. For the propositional function $P(x)$ stating "$x+2 = x$" the existential quantification is "there exists a value of x for which $x+2=x$ is valid". The domain is the set of all real numbers. Obviously the proposition $\exists x, P(x)$ is false because no real value x satisfies the relation $x+2=x$.

Negation of Quantification:

If $P(x)$ be a propositional function on a domain A then it can be proved that.

(i) $\sim(\forall x, P(x)) \equiv \exists x, (\sim P(x))$ for $x \in A$

(ii) $\sim(\exists x, P(x)) \equiv \forall x, (\sim P(x))$ for $x \in A$

Find the truth value of $\forall x, P(x)$ where $P(x)$ is the statement "$3x+1 < 10$" and the domain is the set $\{0, 1, 2, 3\}$.

we see, for $x=3$, $3.3+1 = 10 \not< 10$.

we can't say $3x+1 < 10 \quad \forall x \in \{0, 1, 2, 3\}$

$\therefore$ The proposition $\forall x, P(x)$ is false.

Find the truth value of universal quantifier of the proportional function $P(x, y)$ stating "$x^2+y^2 \leq 12$" and the domain is $\{1, 2, 5\}$

we see for $x=3 \ldots 3^2+3^2 = 18 \not< 12$

we can't say, $x^2+y^2 < 12$ is not valid

$\forall x, y \in \{1, 2, 3\}$

$\therefore$ So, the universal quantifier is false.

Find the truth value of $\forall x \exists y \; P(x, y)$ where $P(x, y)$ states that "$x^2+y^2 < 12$" and the domain is $\{1, 2, 3\}$.

For $x=1, y=1$, $x^2+y^2 = 1+1 = 2 < 12$

For $x=2, y=1$, $x^2+y^2 = 4+1 = 5 \leq 12$

For $x=3, y=1$ $x^2+y^2 = 9+1 = 10 < 12$

so, $\forall x \exists y \; P(x, y)$ where $x, y \in \{1, 2, 3\}$ is true.

Let $P(x)$ be the statement "x can speak english" $Q(x)$ be the statement "x knows hindi".

Translate the following into symbolic statements.

(i) Every student of the school either can speak english or knows hindi.

(ii) There is a student who can't speak english and also do not know hindi.

(i) $\forall x, P(x) \vee Q(x)$.

(ii) $\exists x, \sim P(x) \wedge \sim Q(x)$.

Show that; universal quantifier distributes over conjunction and existential quantifier distributes over disjunction.

we want to show,

$$\forall x, (P(x) \wedge Q(x)) \equiv (\forall x, P(x)) \wedge (\forall x, Q(x)).$$

LHS: for all x, $P(x)$ & $Q(x)$ is true.

RHS: $P(x)$ is true for all x & $Q(x)$ is true for all x.

so. from both sides $P(x)$ & $Q(x)$ are true for every x → They are logically equivalent.

Hence universal quantifier distributes over conjunction.

we want to show.

$$\exists x, (P(x) \vee Q(x)) \equiv (\exists x, P(x)) \wedge (\exists x, Q(x)).$$

LHS : There exists at least one 'x' for which either $P(x)$ or $Q(x)$ is true.

RHS : There exists one 'x' for which $P(x)$ is true Either

or there exist one x for which $Q(x)$ is true.

$\therefore$ At least one of $P(x)$ or $Q(x)$ is true for some x.

Hence, existential quantifier distributes over disjunction.

Suppose $R(x,y)$ is the predicate "x understands y". The universe of discource of x is the set of students in your discrete class, and the universe of discource of y is the set of examples in these lecture notes. Then write the following symbolic expression in words -

1. $\exists x, \forall y \quad R(x,y)$
2. $\forall y \quad \exists x \quad R(x,y)$
3. $\forall x \quad \exists y \quad R(x,y)$
4. $\exists y \quad \forall x \quad R(x,y)$.

$R(x,y)$: x understands y.

x : Students of discrete class.

y : Examples in the lecture notes.

1. There is a student who understands every ~~student~~ examples in the lecture ~~st~~ notes.
2. For every ~~st~~ example in the lecture notes there exist some student who understands it.
3. Every student understands ~~st~~ atleast one example in the lecture note.

4. There is atleast one example in the lecture notes, which every student understands.